AARON LI

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Education

Stanford University | *Master of Science in Mechanical Engineering* — Expected June 2027
GPA: 4.00/4.00; Depth in Mechatronics, Breadth in Design & Manufacturing

University of California, Los Angeles | *Bachelor of Science in Bioengineering* — Conferred June 2025
GPA: 3.95/4.00; *magna cum laude*

Technical Skills

Software/CAD: Autodesk Fusion 360, SolidWorks, Onshape, KiCad, Minitab, C/C++, Python, JavaScript, MATLAB
Manufacturing: Lathe, Mill, Welding, Laser Cutting, FDM & SLA 3D Printing

Engineering Experience

Product Realization Lab (PRL) at Stanford University | Stanford, CA — 09/26 – Present
Course Assistant
- Mentor students in the design, fabrication, and testing of hands-on engineering projects using machining, welding, additive manufacturing, laser cutting, and rapid prototyping tools.
- Provide technical instruction and laboratory support for engineering design courses.

Johnson & Johnson MedTech | Santa Clara, CA — 06/26 – 09/26
R&D Systems Verification Engineering Intern
- Supported development and execution of Verification testing for OTTAVA Surgical Robotic System.
- Developed a Python/YAML automated configuration-management tool for a distributed robotic system and accompanying pytest-based validation framework for 25+ automated test cases.
- Designed and prototyped robotic test fixtures in SolidWorks for system-level verification testing.

Center for Advanced Surgical & Interventional Technology at UCLA | Los Angeles, CA — 03/24 – 06/25
Undergraduate Researcher – Haptic Sensors for Robotic Surgery
- Developed haptic sensors for the da Vinci Surgical Robotic System using FBG sensing to detect tissue stiffness.
- Designed method to measure grasping kinematics from robotic forceps using optical encoders.
- Devised protocol for customized forceps compatibility within da Vinci System for experimental data collection.

Biomedical Engineering Society at UCLA | Los Angeles, CA — 04/23 – 06/25
Technical Project Manager/Lead
- Led 40+ students across different projects focused on smart wearable technology and medical devices.
- Spearheaded workshops on coding, circuitry, computer-aided design, printed circuit board design, and soldering while guiding students in building their own medical devices from scratch.
- Designed and maintained year-long project timelines and curricula while managing $4,500+ in funds.
- Coordinated with faculty and alumni for technical mentorship and timeline execution.

Johnson & Johnson MedTech (Shockwave Medical) | Santa Clara, CA — 06/24 – 08/24
R&D Systems Engineering Intern
- Created and performed product testing for Class II & III medical device within Product Development Process.
- Executed device qualification for Verification & Validation testing through failure analysis of prototype builds.
- Implemented automated defect management analysis tool for device requirements, reducing workload by 50%.

Projects

Autonomous Ball-Manipulating Robot — 2/26
Stanford Smart Product Design Applications (ME218B) Final Project
- Designed and built an autonomous robot system using a PIC32-based circuit with IR sensors, DC motors with encoders, and Servo motors programmed by embedded C to navigate a field and manipulate rubber balls.

Haptic Block Stacking Arcade Game — 05/26 – 06/26
Stanford Design and Control of Haptic Systems (ME327) Final Project
- Designed and manufactured an arcade game that provided haptic feedback when stacking virtual game blocks using DC motors with encoders, ERM motors, linear rail, ATmega328 microcontroller, and Processing IDE.

Leadership and Engagement

China Care Bruins at UCLA | Los Angeles, CA — 09/21 – 06/25
Youth Mentorship Director
- Facilitated mentorship between students and international youth adoptees, fostering an inclusive community.
- Developed accessible activities for all adoptees, including individuals with disabilities and special needs.

Interests: Backpacking, bouldering, landscape photography, and budget traveling